ITVNA0-22-Formative Assessment Project-Mowbray Campus-5737241

Network+ Project 1: Subnetting, Virtualisation & Troubleshooting

Student Name: Ameer Ismail Rasool

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Question 1: Subnetting

[Steps for the video are below]

1.1. Creating a UTP Straight Cable

Tools and Materials Needed:

- UTP (Unshielded Twisted Pair) cable
- RJ-45 connectors
- Crimping tool
- Cable tester (optional)
- Wire cutter/stripper
- Pliers
- Scissor

Steps to Create a UTP Straight Cable:

1) Cut the Cable:

Measure the necessary length of cable and cut it using the wire cutter.

2) Strip the Cable:

Strip about 2cm of the outer insulation from both ends of the cable using the wire stripper. One should be careful not to nick the inner wires.

3) Untwist the Pairs:

Within the cable are four twisted pairs of wires. Untwist each pair and straighten them out.

4) Arrange the Wires:

For a straight-through cable, place the wires in the following order (left to right when the clip is not facing you):

Pin 1: White/Orange

Pin 2: Orange

Pin 3: White/Green

Pin 4: Blue

Pin 5: White/Blue

Pin 6: Green

Pin 7: White/Brown

Pin 8: Brown

5) **Trim the Wires:**

Trim the wires to be the same length, about 2 cm from the end of the insulation.

6) **Insert Wires into RJ-45 Connector:**

Carefully insert the wires into the RJ-45 connector, ensuring they are still in the same order. Gently push the wires all the way in until they reach the end of the connector.

7) **Crimp the Connector:**

Use the crimping tool to crimp the RJ-45 connector onto the cable. This will keep the wires in place.

8) **Test the Cable:**

If you have a cable tester, use it to ensure the cable is functioning well. Plug both ends of the cable into the tester and ensure all lights indicate a good connection.

9) **Repeat for the Other End:**

Repeat steps 2 to 8 for the other end of the cable.

1.2 Perform Network Calculations

a. Calculate the Subnet Mask and Subnet Address (10 Marks)

Identify the CIDR Notation:

The given subnet address is *192.168.1.66/28*.

Calculate the Subnet Mask:

- The "/28" indicates that the first 28 bits are for the network part.

The subnet mask in:

Binary is: *11111.11111111.11111111.11110000*

Decimal is: *255.255.255.240*

Determine the Subnet Address:

- The subnet address is the first address in the subnet.

For a /28 subnet, the block size is 16.

- The subnet address is calculated by rounding down the host address (192.168.1.66) to the nearest multiple of 16: *192.168.1.64*.

b. Calculate the First and Last Host for the Subnet (10 Marks)

First Host Address:

- The first host address is one more than the subnet address:

First Host = 192.168.1.64 + 1 = *192.168.1.65*

Last Host Address:

- The last host address is one less than the broadcast address.
- The total number of addresses in a /28 subnet is 2^(32-28) = 16

Therefore, the last host is:

- Last Host = 192.168.1.64 + 16 - 2 = *192.168.1.78*

c. Calculate the Broadcast Address for the Subnet (5 Marks)

Broadcast Address:

- The broadcast address is the last address in the subnet.
- For a /28 subnet, the broadcast address is:

Broadcast Address = 192.168.1.64 + 16 - 1 = *192.168.1.79*

Calculation	***Result***
Subnet Mask	*255.255.255.240*
Subnet Address	*192.168.1.64*
First Host Address	*192.168.1.65*
Last Host Address	*192.168.1.78*
Broadcast Address	*192.168.1.79*

Question Two: Virtual Machine Configuration Guide

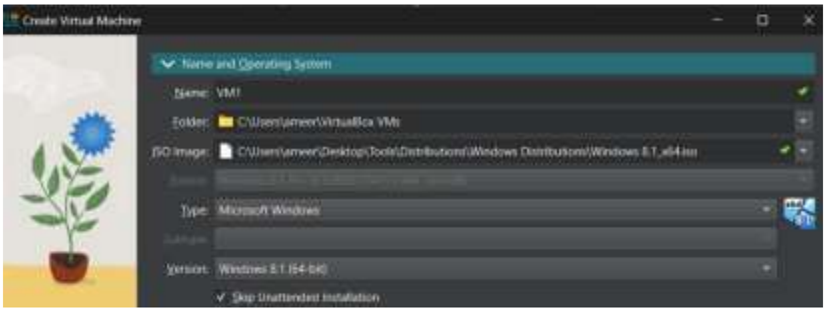
Set Up Virtual Machines (30 Marks)

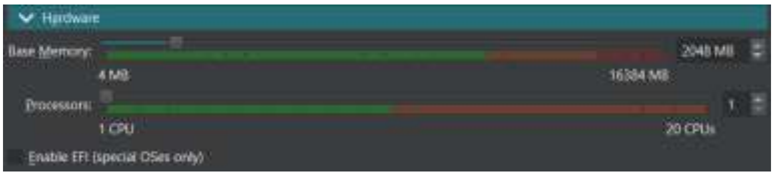
1.1 Create Virtual Machines

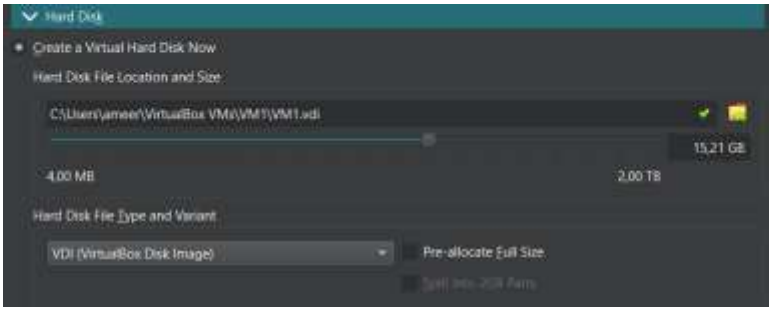
Create VM1 (Windows 8.1)

1. **Open Oracle VirtualBox:** Launch the Oracle VirtualBox application.
2. **Create VM1:** Click "New" to create a new VM.
   Allocate RAM (2048 MB).
   Create virtual hard disk (15 GB). Follow prompts to finish creation.

   Screenshot:





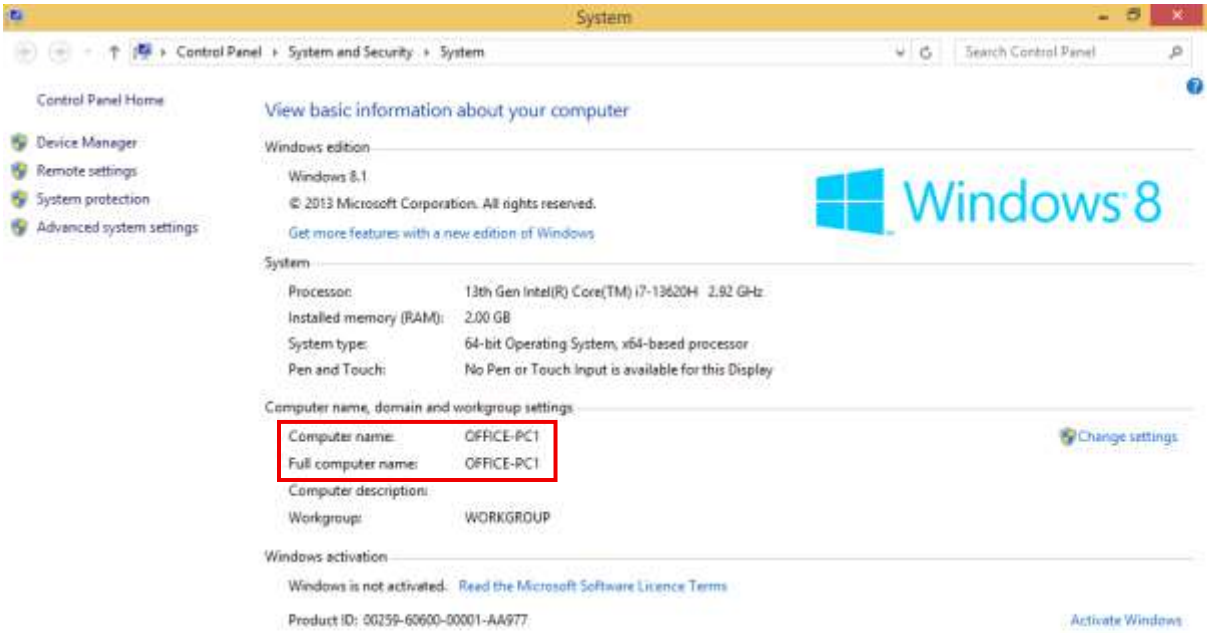


3. **Install Windows 8.1:** Start VM1, select Windows 8.1 ISO, follow installation prompts.

   Screenshot:

4. **Set Computer Name:** Set to *OFFICE-PC1* via: Right-click “This PC” → Properties → Change settings → Change → Enter *OFFICE-PC1*.

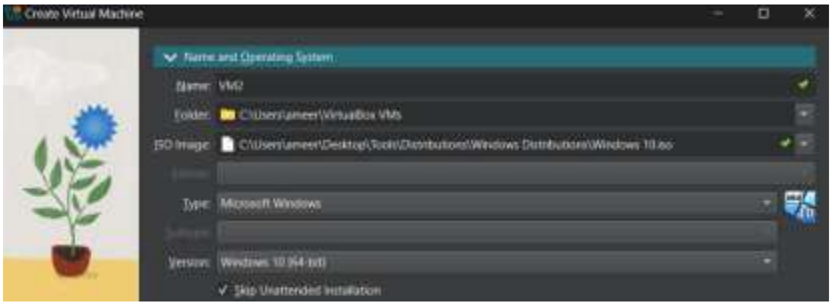
   Screenshot:



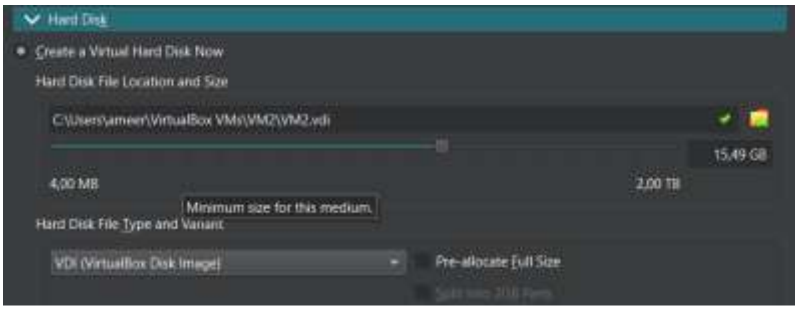
Create VM2 (Windows 10)

1. Repeat VM creation steps for new machine named *VM2*.

   Screenshot:

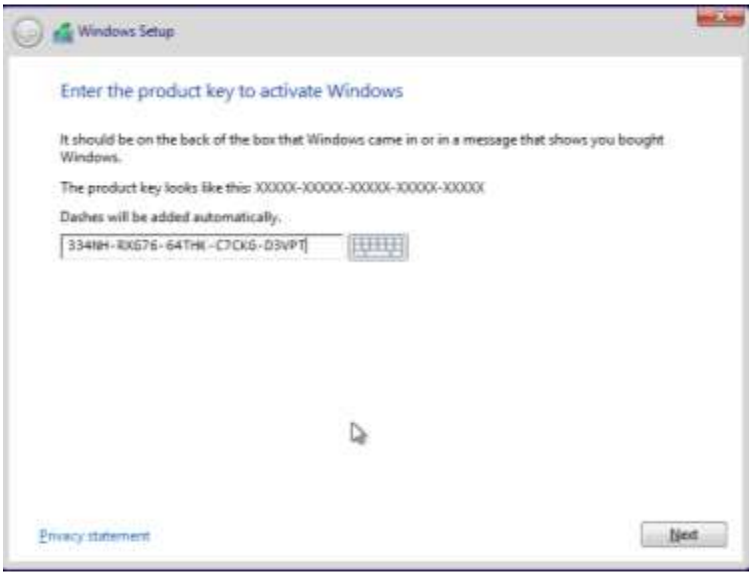


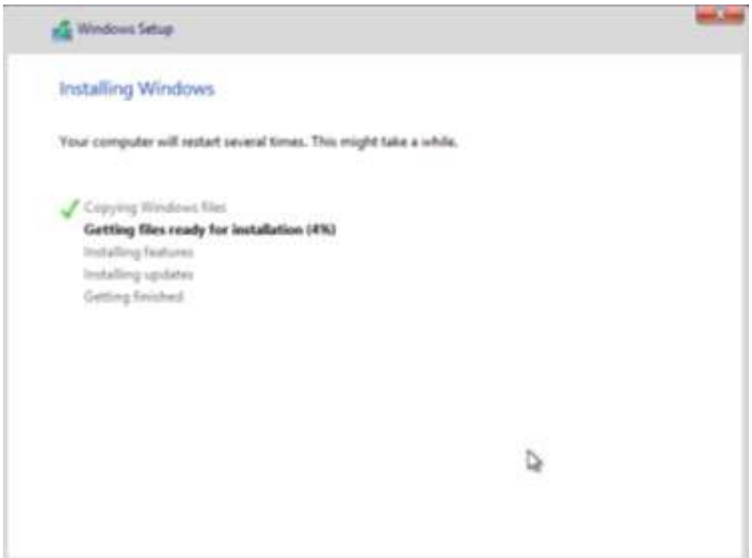
Allocate RAM and virtual hard disk similarly.



2. Install Windows 10 on VM2 via ISO file.

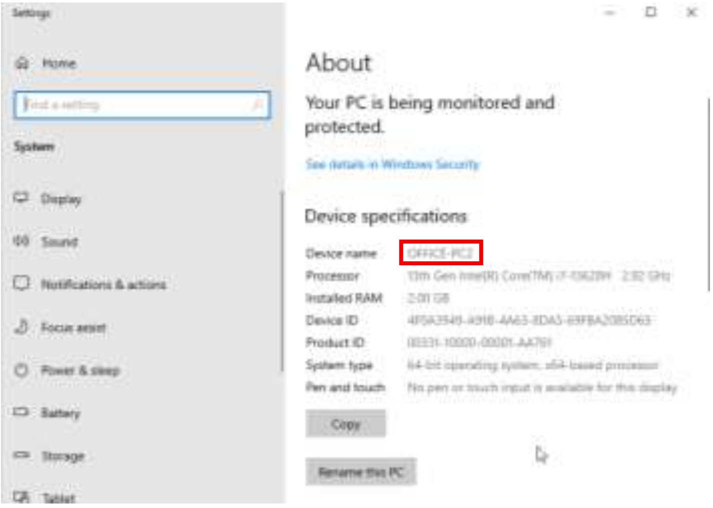
Screenshot:





3. Set computer name to *OFFICE-PC2*.

   Screenshot:



Configure Workgroup (5 Marks)

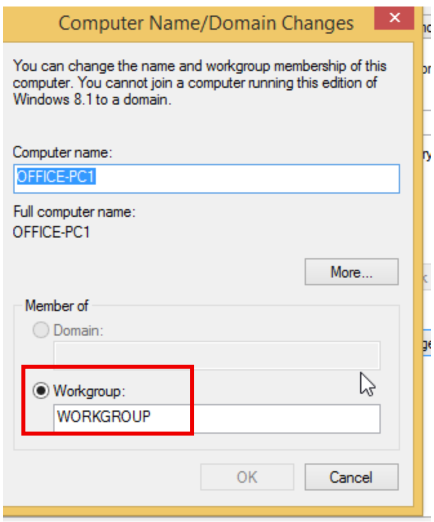
2.1 Set Workgroup for Both PCs

Configure Workgroup on OFFICE-PC1:

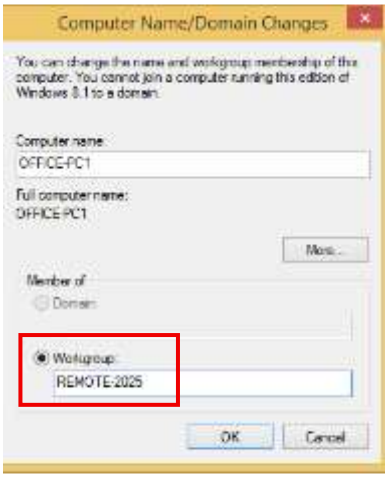
1. Go to Control Panel > System and Security > System on OFFICE-PC1.
2. Click "Change settings" next to computer name.
3. Click "Change" and set Workgroup to *REMOTE-2025*.
4. Click OK and restart the VM.

Screenshot:

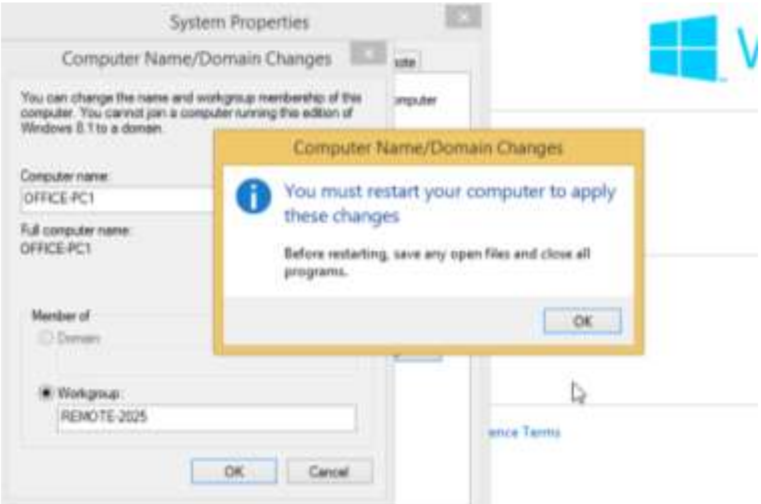
Workgroup before change:



WorkGroup Afterwards:



Note: You will be required to restart the machine before any changes can be applied.

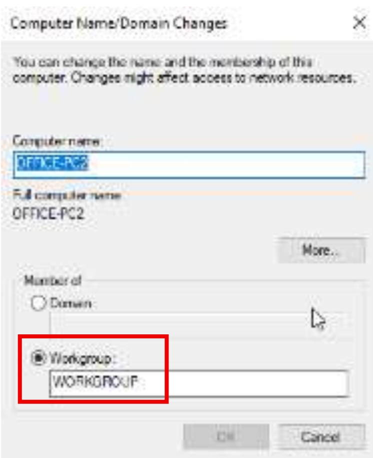


Configure Workgroup on OFFICE-PC2:

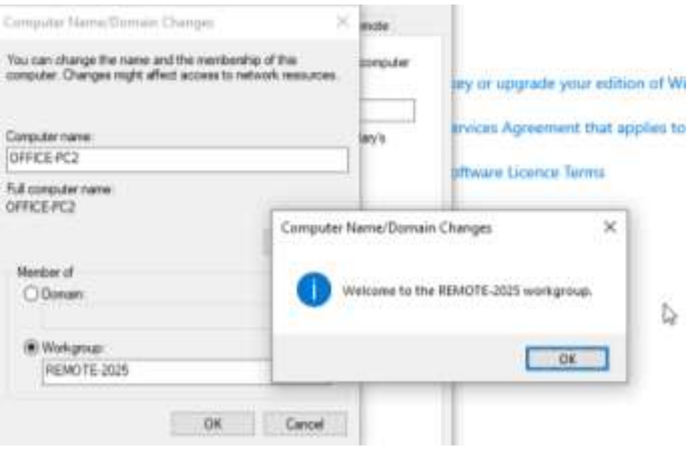
1. Repeat above steps on OFFICE-PC2.

Screenshot:

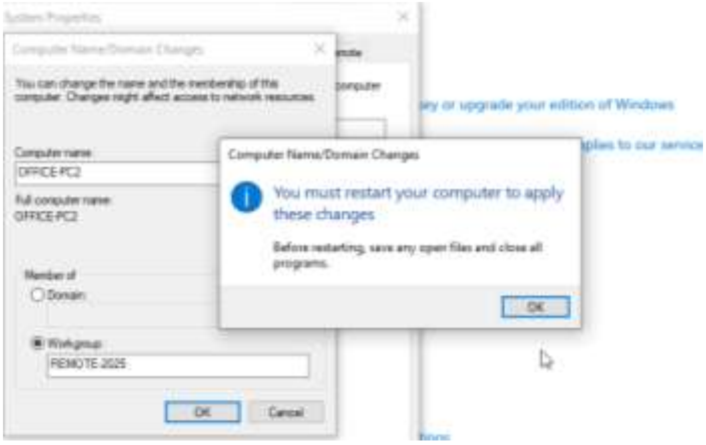
Before changing the Workgroup:



After changing workgroup:



Note: machine must restart before any changes are applied.



Configure IP Addresses (10 Marks)

3.1 Configure IP Addresses

Configure OFFICE-PC1:

1. Control Panel > Network and Internet > Network and Sharing Center > Change adapter settings.
2. Right-click network adapter, select “Properties”.
3. Select “Internet Protocol Version 4 (TCP/IPv4)”, click “Properties”.
4. Set IP to first host address from Question 1.2b (192.168.1.65).
5. Set Subnet Mask to subnet mask from Question 1.2a (255.255.255.240).
6. Click OK to save.

Screenshot:

Configure OFFICE-PC2:

1. Repeat on OFFICE-PC2 with last host IP from Question 1.2b (192.168.1.78).
2. Use same Subnet Mask.

Screenshot:

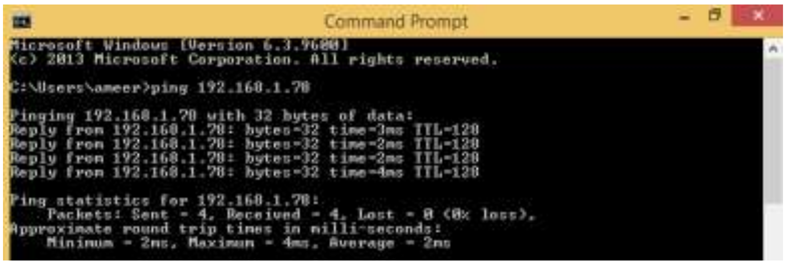
Test Connectivity (5 Marks)

4.1 Ping Between PCs

Ping OFFICE-PC2 from OFFICE-PC1:

1. Open Command Prompt on OFFICE-PC1.
2. Type: ping 192.168.1.78

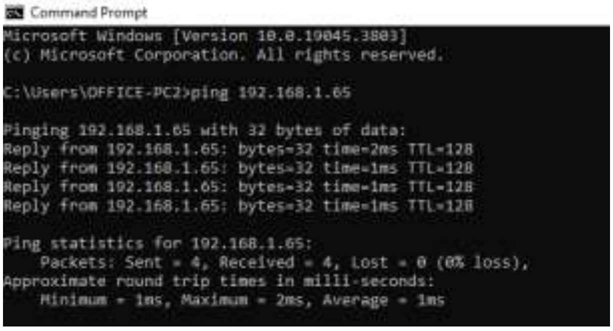
Screenshot:



Ping OFFICE-PC1 from OFFICE-PC2:

1. Open Command Prompt on OFFICE-PC2.
2. Type: ping 192.168.1.65

   Screenshot:



Configure DNS Server (5 Marks)

5.1 Set DNS Server

Configure DNS on OFFICE-PC1:

1. Go to Control Panel > Network and Internet > Network and Sharing Centre > Change adapter settings.
2. Right-click network adapter, select "Properties".
3. Select "Internet Protocol Version 4 (TCP/IPv4)", click "Properties".
4. Choose "Use the following DNS server addresses" and enter Preferred DNS server: *(10.16.0.10)*.
5. Click OK to save.

Screenshot:

Configure DNS on OFFICE-PC2:

1. Repeat above steps on OFFICE-PC2 with the same preferred DNS server.

   Screenshot:

Question 3: Troubleshoot issue with DNS server

Analysis of the Network Connectivity Problem

Problem Description:

- Heinrich can connect to the server through its IP address but cannot connect using its hostname.
- He receives an error message indicating that "The DNS server isn't responding.

Possible Causes

DNS Settings Problems

The DNS settings on Heinrich's system might be incorrect or improperly configured.

There might be no DNS server assigned, or the DNS server specified is down or out of order.

Network Connectivity Problems

There could be generic network connectivity problems affecting DNS queries.

Firewall configurations may be blocking DNS requests or traffic to the DNS server.

Hostname Resolution Issues

The hostname could not be enrolled in the DNS server.

There might be a cache issue with outdated or incorrect mappings retained.

DNS Server Issues

The DNS server itself can be down or in fault.

There could be an overload of the DNS server, and it might not respond.

Recommended Solutions

Verify DNS Settings

Ensure the device is configured with the correct DNS server addresses.

If need be, switch to a public DNS server (e.g., Google DNS: 8.8.8.8 and 8.8.4.4).

Network Connectivity Check

Verify the network connection as a whole by ensuring other devices can connect to the server.

Perform a ping test to the DNS server to validate its accessibility.

Flush DNS Cache

Flush the DNS cache of Heinrich's machine so that stale records don't hamper the resolution process.

Command*: “ipconfig /flushdns”* (in Windows).

Restart DNS Client Service

Restart the DNS client service of Heinrich's machine in order to restart the process of DNS resolution.

Command*: “net stop dnscache”* and then “*net start dnscache*” (in Windows).

Contact Network Administrators

If the issue persists, consult with network administrators to check if there is an issue with the server or DNS misconfigurations in the network.

Check Firewall Settings

Ensure that there are no firewall settings that are hindering DNS requests or the port (UDP port 53) used exclusively by DNS.

By addressing these areas, Heinrich should be able to resolve his connectivity issues when accessing using the hostname.

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https://www.youtube.com/watch?v=2pzF8nAyOGE